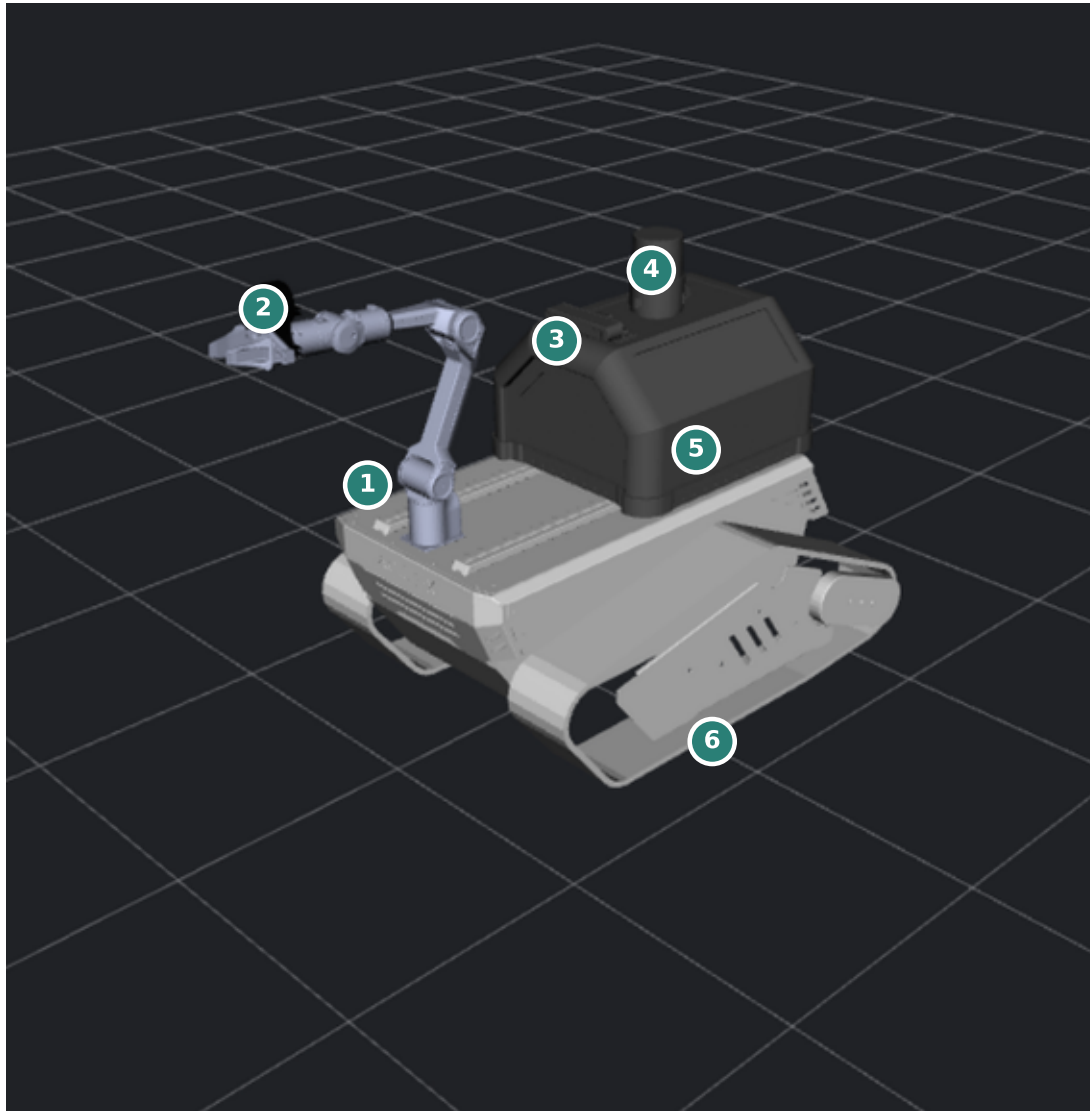


(a) Integrated mobile-manipulation digital model



(b) Hardware roles in the autonomy stack

1

PiPER arm

6-DoF mobile manipulation

2

RealSense L515

wrist-mounted depth camera

3

ZED stereo camera

RGB, pose and depth observations

4

RS-Helios LiDAR

32-beam geometry and obstacle sensing

5

Sensor station

protected sensing / compute enclosure

6

Bunker Pro 2

tracked mobile base

Perception: 3-5 · **Manipulation/mobility:** 1-2, 6